WHY A CONVENTION IS NEEDED

For even-index radicals, more than one real number satisfies the defining equation: both 4 and -4 square to 16. Odd-index radicals never have this problem.

PRINCIPAL ROOT

$$\sqrt[n]{a}$$

split by **whether the index n is even or odd**

2

CASE 1

even index

Convention needed

real roots

the radical symbol denotes the non-negative root

E.G. $\sqrt{16} = 4$, not -4

1

CASE 2

odd index

No convention needed

real root

each real number has exactly one real odd root, so no ambiguity exists

WHEN BOTH ROOTS ARE NEEDED

Write them explicitly: $x^2 = 16 \Rightarrow x = \pm\sqrt{16} = \pm 4$.